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48 years

COURSE-END PROJECT REPORT

On

Brand Popularity and Customer Preference Analysis

Submitted to fulfill the requirements for the Award of CIE Marks (10) for the B.E.(IT), III-SEM,
Course: **Exploratory Data Analysis & Visualization - EDAV (22ADC31N)**

By

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CERTIFICATE

This is to certify that the course-end project report entitled **"Brand Popularity and Customer Preference Analysis"** is a bonafide record of the work carried out by **M. Jeshwanth (1601-25-737-100)**, in partial fulfilment of the requirements of the course **Exploratory Data Analysis and Visualization (22ADC31N)** of the programme **B.E(IT), III-SEM**, during the academic year **2026–2027**.

The project was carried out under my guidance and supervision. To the best of my knowledge, the work reported here is original and has not been submitted to any other institution or university for the award of any degree, diploma or certificate.

Place: Hyderabad
Date: 23/10/2026

Course Faculty
(Prof Ramakrishna Kolikipogu)

DECLARATION

I hereby declare that the project report entitled **"Brand Popularity and Customer Preference Analysis"** submitted to **Dept of IT, CBIT** is a record of original work done by me under the guidance of **Dr Ramakrishna Kolikipogu, Professor, IT, CBIT**.

The data used was taken from the UCI Online Retail dataset (Chen, 2015); all external sources, tools and libraries have been acknowledged in the References. This work has not been submitted, in part or in full, for any other course, degree or diploma.

Place: Hyderabad
Date: 23/10/2026

Signature of the Student
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Abstract

This project applies Python-based exploratory data analysis to real online retail transactions from the UCI Machine Learning Repository. It studies product popularity using quantities and transaction frequency, revenue by product and country, monthly revenue trends, and repeat purchasing among identified customers. Pandas and NumPy are used for data preparation and calculations, while Matplotlib and Seaborn are used for visualizations.

Keywords: online retail; exploratory data analysis; product popularity; revenue analysis; monthly trend; Python; pandas; Seaborn.

1. Introduction

Understanding what customers buy and how purchasing patterns change is useful for business planning. This project applies exploratory data analysis to transaction records to summarize product-level sales, geographic revenue, and repeat-order behavior.

2. Problem Statement and Objectives

Problem statement. Retail transaction data can be large and difficult to interpret manually. This project organizes the data and uses visual and statistical summaries to identify patterns that may inform product and customer analysis.

2.1 Objectives

- Identify products with the highest units sold and transaction frequency.
- Compare product revenue.
- Analyze revenue across countries and months.
- Estimate repeat purchasing among customers with recorded IDs.
- Communicate findings through understandable charts.

3. Dataset Description

The analysis uses the **UCI Online Retail** dataset, created from transactions of a UK-based non-store online retailer between 1 December 2010 and 9 December 2011. The original dataset contains 541,909 rows and fields including invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country. Source: https://archive.ics.uci.edu/dataset/352/online+retail

Table 1: Fields of the dataset

Field	Description
Invoice number	Identifier of each transaction (cancelled invoices are marked)
Stock code	Product code
Product description	Text name of the product (used as a proxy for popularity)
Quantity	Units per product in the transaction
Invoice date	Date and time of the transaction
Unit price	Price per unit in pounds sterling (£)
Customer ID	Identifier of the customer (not recorded for every transaction)
Country	Country of the customer

4. Important Scope Note

The dataset does not provide explicit brand names or customer satisfaction ratings. Therefore, product descriptions are used as a proxy for product popularity, and purchasing behavior is analyzed instead of claiming survey-based customer preference or satisfaction. Sales patterns do not prove why a customer selected a product.

5. Methodology and Tools

The project follows an eight-step pipeline:

1. Load the UCI dataset from a local Excel/CSV file or attempt programmatic retrieval.
2. Inspect columns, missing values, and data types.
3. Parse dates and numeric fields.
4. Exclude cancelled invoices, negative/zero quantities, and non-positive prices from positive-sales analysis.
5. Calculate revenue as Quantity × UnitPrice.
6. Group records by product, country, and month.
7. Create bar charts and a monthly trend line.
8. Save summary tables and metrics for the final findings section.

Table 2: Tools used

Purpose	Tool / Library
Data handling	Python, pandas, NumPy
Charts	Matplotlib, Seaborn
Notebook environment	Jupyter Notebook
Version control	GitHub
Dataset access	openpyxl (reads the Excel file); ucimlrepo (programmatic retrieval attempt)

6. Exploratory Data Analysis and Visualizations

The analysis script generates four charts: top products by units sold, top products by revenue, top countries by revenue, and the monthly revenue trend.

6.1 Top products by units sold

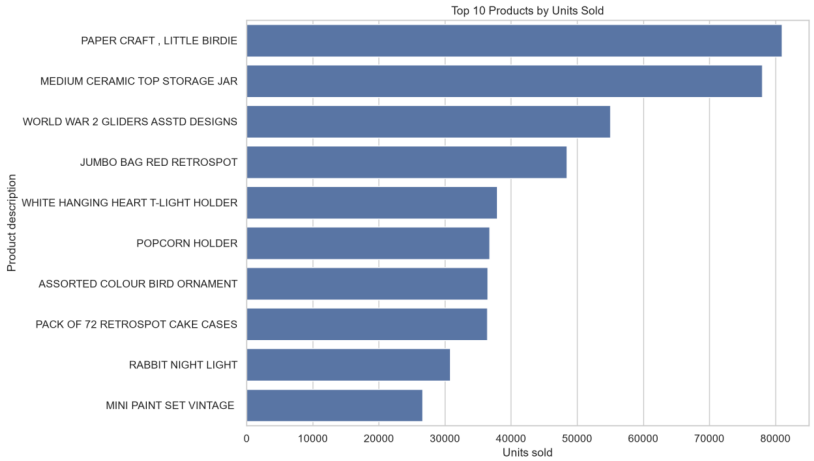


Figure 1: Top 10 products by units sold.

Table 3: Top 10 products by units sold

Rank	Product description	Units sold
1	PAPER CRAFT , LITTLE BIRDIE	80,995
2	MEDIUM CERAMIC TOP STORAGE JAR	78,033
3	WORLD WAR 2 GLIDERS ASSTD DESIGNS	55,047
4	JUMBO BAG RED RETROSPOT	48,474
5	WHITE HANGING HEART T-LIGHT HOLDER	37,891
6	POPCORN HOLDER	36,761
7	ASSORTED COLOUR BIRD ORNAMENT	36,461
8	PACK OF 72 RETROSPOT CAKE CASES	36,419
9	RABBIT NIGHT LIGHT	30,788
10	MINI PAINT SET VINTAGE	26,633

Note: product descriptions are not official brand names, and the dataset has no rating or satisfaction field.

6.2 Top products by revenue

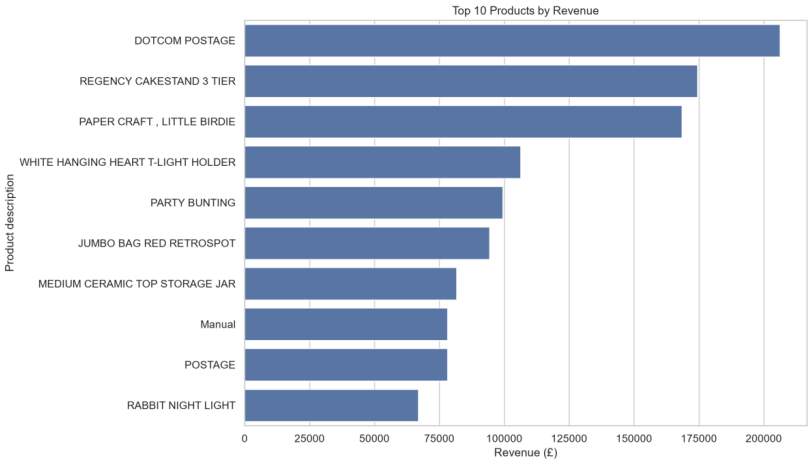


Figure 2: Top 10 products by revenue (£).

Revenue ranks products differently from units sold. DOTCOM POSTAGE, REGENCY CAKESTAND 3 TIER and PAPER CRAFT , LITTLE BIRDIE lead on revenue, followed by WHITE HANGING HEART T-LIGHT HOLDER and PARTY BUNTING. Entries such as DOTCOM POSTAGE, POSTAGE and Manual are service or administrative lines rather than merchandise, so they should be read separately from genuine product popularity.

6.3 Top countries by revenue

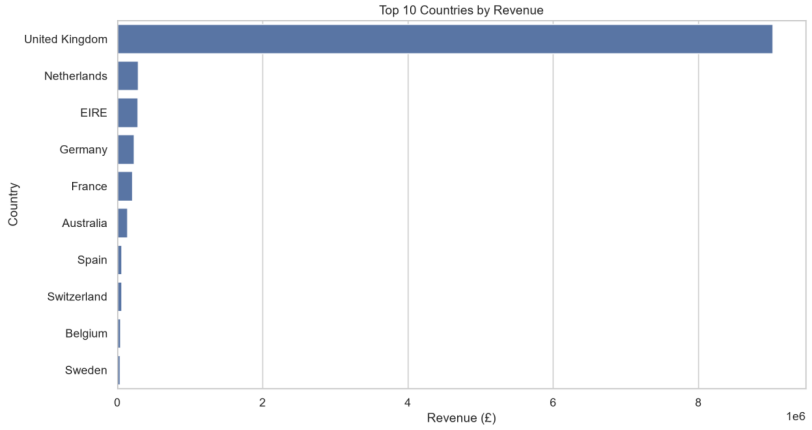


Figure 3: Top 10 countries by revenue (£, scale in millions).

Revenue is overwhelmingly concentrated in the United Kingdom, far ahead of every other market. The Netherlands, EIRE, Germany, France and Australia follow, each contributing only a small fraction of the UK total.

6.4 Monthly revenue trend

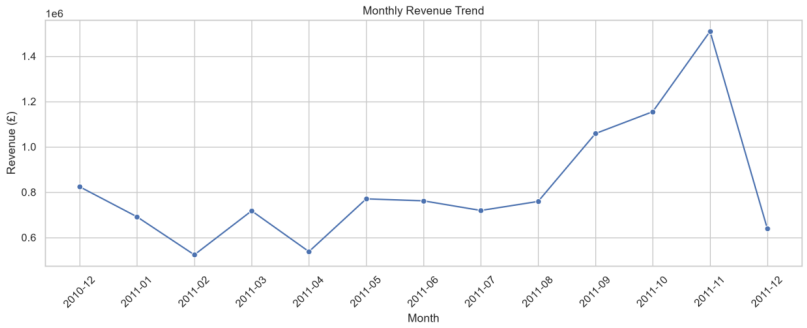


Figure 4: Monthly revenue trend, December 2010 – December 2011 (£, scale in millions).

Monthly revenue is relatively flat from early 2011 to August, then rises sharply through September and October to a peak in November 2011, the strongest month in the period. The apparent fall in December 2011 is expected: the dataset ends on 9 December 2011, so that month is only partially covered.

7. Key Findings

- Units sold and revenue give different rankings. PAPER CRAFT , LITTLE BIRDIE (80,995 units) and MEDIUM CERAMIC TOP STORAGE JAR (78,033 units) lead on volume; DOTCOM POSTAGE and REGENCY CAKESTAND 3 TIER lead on revenue.
- PAPER CRAFT , LITTLE BIRDIE, WHITE HANGING HEART T-LIGHT HOLDER, JUMBO BAG RED RETROSPOT, MEDIUM CERAMIC TOP STORAGE JAR and RABBIT NIGHT LIGHT appear in both the top-10 by units and the top-10 by revenue.
- Several top-revenue entries (DOTCOM POSTAGE, POSTAGE, Manual) are not merchandise.
- The United Kingdom accounts for the large majority of revenue; international markets are comparatively small.
- Revenue is strongly seasonal, peaking in November 2011 after a sharp rise from September.
- December 2011 is a partial month and should not be read as a real decline.

8. Recommendations

- Report units sold and revenue side by side, since each measures popularity differently.
- Exclude postage, manual and other non-merchandise lines when ranking products by popularity.
- Plan stock and promotions ahead of the September–November peak.
- Treat export markets such as the Netherlands, EIRE, Germany and France as growth opportunities, given the heavy reliance on the UK.

9. Limitations

The dataset is historical and relates to one retailer. It does not contain explicit brand labels, customer ratings, stated reasons for purchase, or complete customer identifiers for every transaction. Revenue and quantity are different measures of popularity, and results may not generalize to all markets.

10. Conclusion

This project demonstrates the use of Python for practical exploratory data analysis. It transforms raw retail transactions into summaries and visualizations that reveal product sales patterns, country-level revenue, monthly trends, and repeat purchasing among identified customers. The project can be extended with customer surveys or review data containing actual brands and satisfaction ratings.

11. Future Scope

- Add a survey dataset with brand labels and customer ratings.
- Analyze customer reviews using natural language processing.
- Build an interactive dashboard.
- Compare purchasing patterns over longer time periods.

References

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5. Seaborn documentation: https://seaborn.pydata.org/
6. GitHub documentation: https://docs.github.com/